

INTERNAL ASSESSMENT – ANSWER SCRIPT

1. Subject : Computer Science Fundamentals
2. Course Code : CS-201
3. Semester : III
4. Exam Type : Mid-Semester Internal
5. Total Marks : 11
6. Duration : 30 Minutes
7. Date : 21/05/2026

INSTRUCTIONS FOR THE EVALUATOR

1. Award marks strictly as per the rubric provided for each question.
2. - Evaluate based on conceptual understanding, not exact wording.
3. - Partial marks may be awarded as per rubric breakdown.
4. - Spelling errors should be ignored unless they change the meaning.

Q1. Define a Data Structure. Give one example. [1 mark]

Model Answer:

A data structure is a specialized format for organizing, storing, and managing data in a computer so it can be accessed and modified efficiently. Examples include arrays, linked lists, stacks, and queues.

Rubric:

- Correct definition (organization/storage of data) + at least one valid example (1 mark)

Q2. What is an Algorithm? [1 mark]

Model Answer:

An algorithm is a finite, step-by-step procedure or set of well-defined instructions designed to solve a specific problem or perform a particular task. Every algorithm must terminate after a finite number of steps and produce a correct output.

Rubric:

- Mentions step-by-step procedure to solve a problem AND finite/terminating nature (1 mark)

Q3. Difference between Stack and Queue.

[2 marks]

Model Answer:

A Stack is a linear data structure that follows the LIFO

(Last In First Out) principle, where the most recently inserted element is the first to be removed. Its primary operations are push (insert) and pop (remove).

A Queue follows the FIFO (First In First Out) principle, where the first inserted element is the first to be removed. Its primary operations are enqueue (insert at rear) and dequeue (remove from front).

Stacks are commonly used in function call management and undo operations, while queues are used in CPU scheduling and printer task buffers.

Rubric:

- Correctly explains LIFO with Stack operations (1 mark)
- Correctly explains FIFO with Queue operations (1 mark)

Q4. What is Time Complexity? Explain Big O notation

with two examples.

[3 marks]

Model Answer:

Time complexity is a measure of the amount of time an algorithm takes to run as a function of the size of its input. It helps us analyze and compare the efficiency of algorithms independent of hardware.

Big O notation is the standard mathematical notation used to describe the upper bound (worst-case) growth rate of an algorithm's running time, while ignoring constants and lower-order terms.

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- $O(1)$: Constant time – accessing an array element by index.
- $O(n)$: Linear time – performing a linear search in an unsorted list of n elements.
- $O(n^2)$: Quadratic time – bubble sort or selection sort.

Examples

Rubric:

- Defines time complexity correctly (1 mark)
- Explains Big O as upper bound / worst-case growth (1 mark)
- Provides at least two correct examples with matching complexity classes (1 mark)

Q5. Differentiate between Process and Thread.

Give one real-world example.

[3 marks]

Model Answer:

A Process is an independent program in execution that has its own separate memory space, system resources, and process control block. Processes are isolated from each other, so inter-process communication is slower and more complex.

A Thread is the smallest unit of execution within a process. Multiple threads share the same memory space and resources of their parent process, which makes thread creation lightweight and inter-thread communication fast.

Real-world example: A web browser is a process. Each open tab within the browser is typically a separate thread (or sometimes a separate process for security), allowing tabs to run concurrently and share the browser's resources.

Rubric:

- Correctly defines Process with memory/isolation point (1 mark)
- Correctly defines Thread with shared memory point (1 mark)
- Provides a clear and correct real-world example (1 mark)

END OF ANSWER SCRIPT
